

TRIGONOMETRIC IDENTITIES AND EQUATIONS – PRACTICE PROBLEMS

1. The general value of θ satisfying the equation $\tan^2\theta + \sec 2\theta = 1$ is -

- (a) $m\pi, n\pi + \pi/3$
- (b) $m\pi, n\pi \pm \pi/3$
- (c) $m\pi, n\pi \pm \pi/6$
- (d) none of these

2. $\sin 6\theta + \sin 4\theta + \sin 2\theta = 0$, then θ equal to -

- (a) $n\pi/4$ or $n\pi \pm \pi/3$
- (b) $n\pi/4$ or $n\pi \pm \pi/6$
- (c) $n\pi/4$ or $2n\pi \pm \pi/6$
- (d) None of these

3. The number of solutions of the equation $\tan x + \sec x = 2 \cos x$ lying in the interval $[0, 2\pi]$ is

- (a) 3
- (b) 2
- (c) 1
- (d) 0

4. The general solution of the equation $\sin^{50}x - \cos^{50}x = 1$

- (a) $2n\pi + \pi/2$
- (b) $2n\pi + \pi/3$
- (c) $n\pi + \pi/2$
- (d) $n\pi + \pi/3$

5. Let $2\sin^2x + 3\sin x - 2 > 0$ and $x^2 - x - 2 < 0$ (x is measured in radians). Then x lies in the interval

(a) $(\pi/6, 5\pi/6)$

(b) $(-1, 5\pi/6)$

(c) $(-1, 2)$

(d) $(\pi/6, 2)$

6. The equation $2\cos^2(x/2) \cdot \sin^2 x = x^2 + 1/x^2$, $0 \leq x \leq \pi/2$ has

(a) one real solution

(b) no solution

(c) more than one real solution

(d) none of these

7. If $\sin^4 x + \cos^4 y + 2 = 4 \sin x \cos y$ and $0 \leq x, y \leq \pi/2$ then $\sin x + \cos y$ is equal to

(a) -2

(b) 0

(c) 2

(d) none of these

8. If $0 \leq x \leq 2\pi$ and $|\cos x| \leq \sin x$ then

(a) $x \in [0, \pi/4]$

(b) $x \in [\pi/4, \pi/2]$

(c) $x \in [\pi/4, \pi/2] \cup [\pi/2, 3\pi/4]$

(d) None of these

9. If $\left(\frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta - \cos \theta}\right) - \frac{\cos \theta}{\sqrt{1 + \cot^2 \theta}} - 2 \tan \theta \cot \theta = -1$

$\theta \in [0, 2\pi]$ then

(a) $\theta \in (0, \pi/2] - \{\pi/4\}$

(b) $\theta \in (\pi/2, \pi) - \{3\pi/4\}$

(c) $\theta \in (\pi, 3\pi/2) - \{5\pi/4\}$

(d) $\theta \in (0, \pi) - \{\pi/4, \pi/2\}$

10. Value of $2^{\sin x} + 2^{\cos x}$ is

(a) $\geq 2^{1-\frac{1}{\sqrt{2}}}$

(b) $\leq 2^{1-\frac{1}{\sqrt{2}}}$

(c) $\geq 2^{1+\frac{1}{\sqrt{2}}}$

(d) $\leq 2^{1+\frac{1}{\sqrt{2}}}$

11. Find the solution of the equation $\cos^2 x - 2\cos x = 4\sin x - \sin 2x$, ($0 \leq x \leq \pi$).

12. Find the general solution of equation $\cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta = 0$.

13. Find the general value of θ satisfying the equation $3\tan(\theta - 15^\circ) = \tan(\theta + 15^\circ)$.

14. Find the value of θ satisfying $\cos \theta + \sqrt{3} \sin \theta = 2$.

15. Find the value of θ satisfying $\sec \theta - 1 = (\sqrt{2} - 1)\tan \theta$.

16. $\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ$ is equal to

(a) $\sqrt{3}/4$

(b) $\sqrt{3}/2$

(c) $\sqrt{3}$

(d) 1

17. If $3\tan \theta \tan \phi = 1$, then $\cos(\theta - \phi) / \cos(\theta + \phi) =$

(a) $1/2$

(b) 2

(c) $1/3$

(d) 3

18. The value of $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ$ is

(a) 4

(b) 12

(c) -2

(d) None of these

19. If $\tan\alpha = 1/3$ and $\tan(\beta/2) = 1/2$, then $\tan(\alpha + \beta) =$

(a) 1

(b) -1

(c) 2

(d) 3

20. If $\alpha + \beta = 90^\circ$, then the maximum value of $\sin\alpha \sin\beta$ is

(a) 1

(b) $1/2$

(c) $3/2$

(d) None of these

21. The expression $(\cos 6x + 6\cos 4x + 15\cos 2x + 10)/(\cos 5x + 5\cos 3x + 10\cos x)$ is equal to

(a) $\cos^2 x$

(b) $1 + \cos x$

(c) $\cos 2x$

(d) $2\cos x$

22. If $x = y \cos(2\pi/3) = z \cos(4\pi/3)$ then $xy + yz + zx =$

(a) -1

(b) 0

(c) 1

(d) 2

23. If A & B satisfying the equation $3\sin^2A + 2\sin^2B = 1$ and $3\sin 2A - 2\sin 2B = 0$, then $A + 2B$ is equal to

- (a) π
- (b) $\pi/2$
- (c) $\pi/4$
- (d) 2π

24. If $(1 + \sin 2x)/(1 - \sin 2x) = \cot^2(a + x)$, $\forall x \in \mathbb{R} - (n\pi + \pi/4)$, $n \in \mathbb{N}$. Then 'a' equal to

- (a) $\pi/4$
- (b) $\pi/2$
- (c) $3\pi/4$
- (d) none of these

25. The sum of the series $\sin\theta.\sec 3\theta + \sin 3\theta.\sec 3^2\theta + \sin^3\theta.\sec 3^3\theta + \dots$ upto n terms.

- (a) $1/2 [\tan 3^n\theta - \tan 3^{n-1}\theta]$
- (b) $[\tan 3^n\theta - \tan \theta]$
- (c) $1/2 [\tan 3^n\theta - \tan \theta]$
- (d) none of these

26. Consider an equilateral triangle with sides = 3 cm. Now, if a man runs around the triangle in such a way that he is always at a distance of 1 cm from the sides of triangle then find the distance covered by him.

27. Prove that $\tan 3A \tan 2A \tan A = \tan 3A - \tan 2A - \tan A$

28. Prove that $(\sec 8A - 1)/(\sec 4A - 1) = \tan 8A/\tan 2A$.

29. Find the value of the expression $16(1 + \cos \pi/8)(1 + \cos 3\pi/8)(1 + \cos 5\pi/8)(1 + \cos 7\pi/8)$.

30. Prove that $\cos A \cos(60^\circ - A) \cos(60^\circ + A) = 1/4 \cos 3A$

31. Prove that $\tan 20^\circ \tan 40^\circ \tan 60^\circ \tan 80^\circ = 3$

32. If $\cos^3 2\theta + 3\cos 2\theta = k(\cos^6 \theta - \sin^6 \theta)$ then find the value of k .

33. If $A + B + C = 180^\circ$, prove that $\sin(A/2) + \sin(B/2) + \sin(C/2) - 1 = 4 \sin((\pi - A)/4) \sin((\pi - B)/4) \sin((\pi - C)/4)$

34. Find the greatest and least value of $\frac{1}{4} \sin^2 \left(\frac{x}{3} - \frac{\pi}{6} \right) - \left(\frac{1}{3} \right) \sin \left(\frac{x}{3} - \frac{\pi}{6} \right) + 1$